An odd-divisor inverse for a shifted Lambert factorization

A counterexample and corrected formula for MathDB 356220

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Abstract

The proposed inverse in Schmidt's Conjecture 2.26 fails at $(n,k) = (58,1)$: it gives 613766, whereas the inverse entry is 613767. Ordinary Möbius inversion on odd integers gives a corrected formula for every entry and every invertible formal prefactor. This is an authoring-agent checked manuscript, without independent review. Campaign credit is zero pending inspection of the related final journal article.

Task 1. Refutation and complete corrected inverse

Let $p(j)$ be the ordinary partition number, with $p(0) = 1$ and $p(j) = 0$ for negative integers j. Define $S = (s_{n,k})_{n,k\geq 1}$ by

$$\sum_{k\geq 1} \frac{f(k)q^k}{1-q^{2k+1}} = \frac{1}{(q;q)_\infty} \sum_{n\geq 1} \sum_{k=1}^{n} s_{n,k} f(k) q^n, \qquad (q;q)_\infty = \prod_{j\geq 1}(1-q^j). \tag{1}$$

These are formal power series; every coefficient uses finitely many terms. The matrix S is unit lower triangular, and $s_{n,k}^{(-1)}$ denotes the entry of its inverse.

Conjecture 2.26 of the thesis [1], printed page 31, proposes

$$\begin{aligned} s_{n,k}^{(-1)} \stackrel{?}{=} p(n-k) &- \sum_{\substack{1\leq i\leq n \\ n\equiv i \ (2i+1)}} p\left(\frac{n-i}{2i+1} - k\right) \\ &+ \sum_{m=2}^{n} \sum_{\substack{1\leq i\leq n \\ n\equiv p(m+1)i+p(m-1) \ (p(m+1)(2i+1))}} p\left(\frac{n-p(m+1)i-p(m-1)}{p(m+1)(2i+1)} - k\right). \end{aligned} \tag{2}$$

The congruences in the summation conditions have the same meaning as the source's Iverson factors. This formula also appears as equation (5) in Remark 2.7 of [2]. The function p in the source is the partition function throughout.

Theorem 1. *For all integers $n \geq k \geq 1$, the actual inverse is*

$$\boxed{s_{n,k}^{(-1)} = \sum_{d\mid 2n+1} \mu(d)\, p\left(\frac{(2n+1)/d - 1}{2} - k\right).} \tag{3}$$

Here μ is the classical Möbius function. In particular $s_{58,1}^{(-1)} = 613767$, while the right side of (2) is 613766. Thus the conjecture is false.

Inversion of the odd-divisor kernel

Write $C(q) = \sum_{j\ge 0} c_j q^j$, where $c_0 = 1$, and $C(q)^{-1} = \sum_{j\ge 0} r_j q^j$, with $r_j = 0$ when $j < 0$. Replacing $(q;q)_\infty$ by $C(q)$ in (1) defines a matrix S_C. Let $T_C = (c_{n-k})_{n,k\ge 1}$, so that $T_C^{-1} = (r_{n-k})_{n,k\ge 1}$.

Lemma 1. *The coefficient matrix of $q^k/(1-q^{2k+1})$ is*

$$D_{n,k} = \mathbf{1}_{\,2k+1\mid 2n+1} \qquad (n,k \ge 1).$$

Its inverse is

$$E_{n,k} = \begin{cases} \mu\big((2n+1)/(2k+1)\big), & 2k+1 \mid 2n+1, \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. The coefficient in question is 1 precisely when $n = k + \ell(2k+1)$ for an integer $\ell \ge 0$. This is equivalent to $2n+1 = (2\ell+1)(2k+1)$. Conversely, the quotient of two odd integers is odd, so divisibility gives such an ℓ.

If $2k+1$ does not divide $2n+1$, the (n,k) entries of both DE and ED vanish. Otherwise put $M = (2n+1)/(2k+1)$. The intermediate odd integers $2j+1$ are exactly $(2k+1)e$ with $e \mid M$. Consequently each product entry is $\sum_{e\mid M} \mu(e)$, which is 1 for $n = k$ and 0 otherwise. The divisor identity follows directly from $\prod_{\ell\mid M,\ \ell \text{ prime}}(1-1)$ when $M > 1$. There is no missing index at the odd integer 1: every intermediate integer is at least $2k+1 \ge 3$. □

Proof of the all-entry formula. Multiplication by $C(q)$ gives $S_C = T_C D$. Thus

$$S_C^{-1} = E T_C^{-1}, \qquad (S_C^{-1})_{n,k} = \sum_{\substack{k\le j\le n \\ 2j+1\mid 2n+1}} \mu\left(\frac{2n+1}{2j+1}\right) r_{j-k}.$$

Substitute $d = (2n+1)/(2j+1)$. This proves

$$(S_C^{-1})_{n,k} = \sum_{d\mid 2n+1} \mu(d)\, r_{\frac{(2n+1)/d-1}{2}-k}. \tag{4}$$

All omitted indices have negative subscripts and therefore contribute zero. For $C(q) = (q;q)_\infty$ we have $r_j = p(j)$, proving (3). All matrix products are finite at each entry. In particular, inversion of the leading N by N block agrees with this formula whenever $n \le N$. □

Exact evaluation at $(58, 1)$

Since $117 = 3^2 \cdot 13$, the nonzero Möbius terms have $d = 1, 3, 13, 39$. Equation (3) gives

$$p(57) - p(18) - p(3) + p(0) = 614154 - 385 - 3 + 1 = 613767.$$

For the negative sum in (2), divisibility gives the following and only the following nonzero terms:

i	1	4	6	19
$(58-i)/(2i+1) - 1$	18	5	3	0

For the positive double sum, a nonzero term requires

$$58 \ge p(m+1)(3i+1) + p(m-1).$$

For $m \ge 6$ the right side is at least $4p(7) + p(5) = 67$. For $m = 2, 3, 4, 5$, the possible i from this bound are respectively $1, \dots, 6$; $1, \dots, 3$; $1, 2$; and 1. Testing the indicated congruences leaves just $(m, i) = (2, 1), (2, 6)$, with partition indices $5, 0$. Hence the proposed value is

$$p(57) - \big(p(18) + p(5) + p(3) + p(0)\big) + \big(p(5) + p(0)\big) = 614154 - 396 + 8 = 613766.$$

This is a failure of the mathematical formula repeated in both sources. The corrected theorem answers the inverse-matrix question for all entries.

Verification and source limits

The accompanying standard-library Python script builds S directly from finite products and geometric series, then inverts it by exact triangular recursion. It checks all 1711 entries through row 58, including the first failure of (2); it also tests (4) for two other prefactors. These finite tests supplement the proof above. Classical Möbius inversion and formal-series matrix multiplication are credited standard tools, not new general theorems.

The complete 2022 thesis and 2017 arXiv preprint were inspected. The final journal version of [2], published in 2020, was not accessible in full. The literature and priority check therefore remains pending, with zero campaign credit. This limitation concerns eligibility, not the exact discrepancy for the displayed formula.

Four proposed follow-up tasks

Task 2. General integer slopes. For each integer $\alpha \geq 3$, obtain an explicit arithmetic inverse for the coefficient matrix of $q^k/(1-q^{\alpha k+1})$. Identify the appropriate divisor-poset Möbius function and its support.

Task 3. The weighted denominator. Determine the inverse entries for $q^k/(1-zq^{2k+1})$ as polynomials in z, and give a factorization-chain interpretation for their coefficients.

Task 4. Positivity of the corrected entries. Determine whether the numbers in (3) are nonnegative for every $n \geq k \geq 1$. If they are, seek a partition interpretation of the odd-divisor inclusion-exclusion sum.

Task 5. Classify the erroneous formula's agreement set. Characterize the pairs (n,k) for which (2) equals (3), and determine the density or growth of rows containing at least one disagreement.

References

[1] M. D. Schmidt, *Factorization theorems and canonical representations for generating functions of special sums*, Ph.D. thesis, Georgia Institute of Technology, 2022, Conjecture 2.26, p. 31. arXiv:2209.12287.

[2] M. Merca and M. D. Schmidt, *Factorization theorems for generalized Lambert series and applications*, arXiv:1712.00611, Remark 2.7, equation (5), pp. 8–9. Published as *Ramanujan J.* **51** (2020), 391–419, doi:10.1007/s11139-018-0095-7. The full published version was not inspected.